Reproducing the Appearance of Metallic Materials by Capturing Long-Range Dependencies using Image-to-Image Translation Network with Vision Transformer

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Abstract. Seamlessly integrating virtual objects into real scenes is a critical challenge in augmented reality. To achieve this, it is important to provide globally consistent estimations of lighting conditions in the real scene. In this study, we propose a method to reproduce the appearance of metallic materials using an image-to-image translation network incorporating Vision Transformer (ViT). The network receives an image consisting of a background and a normal map of a virtual object, and transforms the normal map into a virtual object with the metallic material appearance. Specifically, ViT, which effectively captures long-range dependencies in the input image, is introduced into the network's encoder to extract lighting information, and the network produces the natural appearance of the virtual object that reflects the extracted lighting information in the output image. We created a synthetic dataset and compared images generated by the proposed method with those generated by a CNN-based image-to-image translation network. The results showed that the proposed method outperformed the CNN-based method in three quantitative metrics and reproduced a more natural appearance of metallic materials that is consistent with the lighting conditions.

Keywords: Material Appearance · Illumination Consistency · Image-to-Image Translation · Generative Adversarial Network · Vision Transformer

1 Introduction

In augmented reality (AR), reproducing natural shadows and reflections based on the lighting information of the real environment enhances the user's sense of immersion. In particular, when the virtual object is metallic, it is important

to accurately reproduce the reflection of the surrounding environment on its surface.

There are studies on light estimation in the real environment to enable natural integration of virtual objects into real scenes. Methods were proposed to estimate the lighting information of a real environment and acquire an environment map by using a specular sphere [2, 18, 11] and a camera with a fisheye lens [15, 4]. However, preparing and installing such dedicated equipment is time-consuming because it needs to be deployed in the real world. Some studies estimated lighting information from a single image and generated realistic shadows and reflections for superimposed virtual objects. The high dynamic range environment maps with lighting information of the real environment were acquired by using DNNs to estimate various parameters from a single low dynamic range image for the indoor scene [5, 20] and the outdoor scene [7]. However, since the lighting information of the real environment is estimated using low-dimensional illumination parameters, the reflection of the rendered virtual objects of metallic materials lacks sharpness.

On the other hand, some studies generated natural shadows and reflections for virtual objects in a single image using an image-to-image translation network [17, 19]. In these studies, convolutional neural network (CNN)-based image-to-image translation networks with generative adversarial network (GAN) were used to generate images that reproduced realistic diffuse interreflection and shadows by learning lighting conditions in real environments based on large amounts of training data. However, CNN-based models are less efficient at capturing the long-range dependencies necessary to reproduce reflections of the surrounding environment, such as metals, and may have difficulty generating images under specific environmental conditions. Recently proposed Vision Transformer (ViT) [3] is an architecture that efficiently captures relationships between extensive regions in an image using a self-attention mechanism, achieving state-of-the-art performance on several image recognition benchmarks. ViT architecture has been applied to image generation and image-to-image translation in some cases [9, 6, 1].

In this study, we propose a method to generate natural reflections and shadows of virtual objects in a single image using an image-to-image translation network incorporating ViT into the encoder in the generator. The network transforms an image with a normal map of a virtual object superimposed on a real scene into an image that reproduces the appearance of metallic materials by generating reflections from the surrounding environment. By utilizing the self-attention mechanism of ViT, the network efficiently captures long-range dependencies in the input image and reflects them to the output image, generating the appearance of the metal material consistent with the surrounding environment.

2 Method

We propose a method to reproduce the appearance of metallic materials based on the estimated lighting information from a single image by using an image-to-

image translation network incorporating ViT, which effectively captures long-range dependencies.

2.1 Overview of the network

The overall structure of the network is shown in Fig. 1. The network is based on Vit-GAN [6], which incorporates ViT into pix2pix [8], and we further modified its generator. pix2pix is a type of conditional GAN consisting of a generator and a discriminator, which learns correspondences from input images to target images.

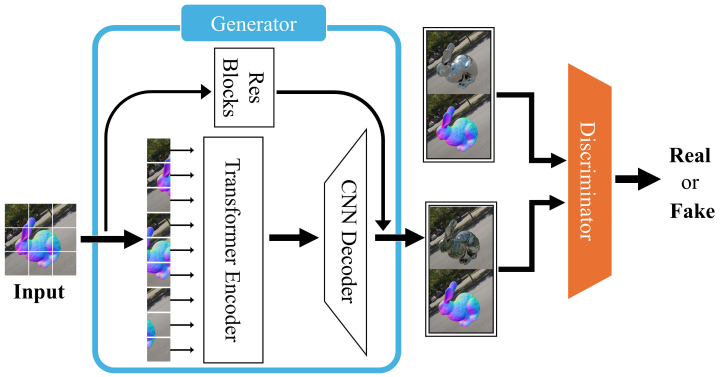

Fig. 1: Overall structure of the network

The network generator consists of a Transformer encoder, a CNN decoder, and several ResBlocks. It is important to capture long-range light transport on a single image to generate the natural appearance of metallic materials. Therefore, we introduce ViT, which efficiently captures long-range dependencies, into the encoder. To generate a natural appearance, it is essential to preserve local features such as strong highlights and sudden changes in brightness and darkness during the upsampling process, which outputs an image with the same resolution as the input image. Therefore, we introduce a CNN-based architecture, which efficiently captures local patterns, into the decoder. Since ViT cannot efficiently model the local structure, such as edges and lines, we introduce ResBlocks with a skip connection so that the structural information of the input image is efficiently conveyed to the output image.

2.2 Network architecture

The structure of the proposed generator is shown in Fig. 2. In the generator, the input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ is first divided into $N = (H \times W)/P^2$ patches of size

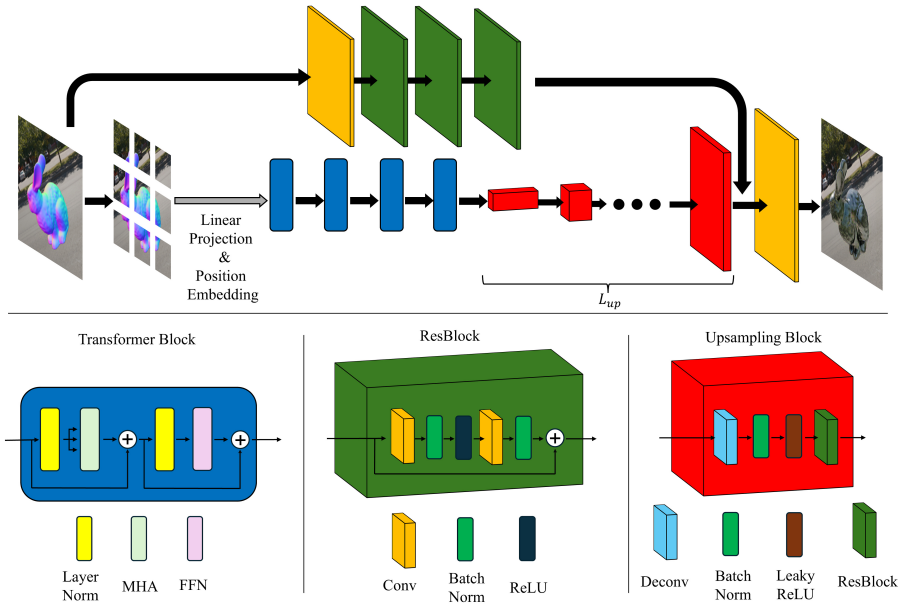

Fig. 2: Structure of the generator network incorporating ViT

$P \times P$, and flattened patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$ are obtained. A linear projection $\mathbf{E}$ using a fully connected layer is applied to the patches, converting them into d-dimensional patch embeddings $\mathbf{z}_p \in \mathbb{R}^{N \times d}$.

$$\mathbf{z}_p = \mathbf{x}_p \mathbf{E} \qquad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times d} \tag{1}$$

Vectors representing the position indices of patches are denoted as position embeddings $\mathbf{E}_p$. $\mathbf{E}_p$ are added to $\mathbf{z}_p$ to obtain embedding vectors $\mathbf{z}'_p \in \mathbb{R}^{N \times d}$.

$$\mathbf{z}'_p = \mathbf{z}_p + \mathbf{E}_p \qquad \mathbf{E}_p \in \mathbb{R}^{N \times d} \tag{2}$$

The resulting embedding vectors $\mathbf{z}'_p$ are subsequently fed into four Transformer blocks. Each transformer block consists of a multihead attention (MHA), a layer normalization (Layer Norm), and a feed-forward network (FFN), similar to ViT's encoder; the FFN consists of two fully connected layers and one ReLU. Given $\mathbf{z}^{(0)} = \mathbf{z}'_p$, the processing of each layer is defined as follows

$$\hat{\mathbf{z}}^{(l)} = \mathrm{MHA}(\mathrm{LayerNorm}(\mathbf{z}^{(l-1)})) + \mathbf{z}^{(l-1)} \qquad l = 1, ..., 4 \tag{3}$$

$$\mathbf{z}^{(l)} = \mathrm{FFN}(\mathrm{LayerNorm}(\hat{\mathbf{z}}^{(l)})) + \hat{\mathbf{z}}^{(l)} \qquad l = 1, ..., 4 \tag{4}$$

where $\hat{\mathbf{z}}^{(l)}$ is the output of the MHA with a skip connection in the lth transformer block, and $\mathbf{z}^{(l)}$ is the output of the FFN with a skip connection in the lth transformer block. The output of fourth transformer block, $\mathbf{z}^{(4)} \in \mathbb{R}^{N\times d}$, is converted into a feature map $\mathbf{I}' \in \mathbb{R}^{N\times\sqrt{d}\times\sqrt{d}}$ by reshaping it. After that, $\mathbf{I}'$ is fed into an upsampling block, and the upsampling blocks are iteratively applied to $\mathbf{I}'$ until its resolution matches that of the input image. Each upsampling block consists of an inverse convolutional layer, a batch normalization, a Leaky ReLU, and a ResBlock. In each upsampling block, the spatial dimensions of the feature map are increased by a factor of 2 in height and width, respectively, and the number of channels is reduced by a factor of 2, for a total of $L_{up} = \log_2(H/\sqrt{d})$ or $L_{up} = \log_2(W/\sqrt{d})$ upsampling blocks. Given that $\mathbf{I}_{\text{up}}^{(0)} = \mathbf{I}'$, the processing in each block is defined by Eq. (5).

$$\begin{aligned} \mathbf{I}_{\text{up}}^{(l')} &= \text{Upsample}(\mathbf{I}_{\text{up}}^{(l'-1)}) \\ l' &= 1, ..., L_{up} \end{aligned} \tag{5}$$

$\mathbf{I}_{\text{up}}^{(l')}$ denotes the output of the l'th upsampling block.

In parallel, a convolutional layer and three ResBlocks are applied to $\mathbf{I}$. The ResBlock consists of a convolutional layer, a batch normalization, and a ReLU. Given that $\mathbf{I}_{\text{res}}^{(0)} = \text{Conv}(\mathbf{I})$, the processing in each block is defined by Eq. (6).

$$\begin{aligned} \mathbf{I}_{\text{res}}^{(l'')} &= \text{ResBlock}(\mathbf{I}_{\text{res}}^{(l''-1)}) \\ l'' &= 1, ..., 3 \end{aligned} \tag{6}$$

$\mathbf{I}_{\text{res}}^{(l'')}$ denotes the output of the l''th ResBlock. Finally, a convolutional layer is applied to the features, which are the sum of the final outputs of the upsampling block and the ResBlock, to match the number of channels in them to that of the input image, and $\mathbf{I}_{\text{out}} \in \mathbb{R}^{H\times W\times C}$ is output.

$$\mathbf{I}_{\text{out}} = \text{Conv}(\mathbf{I}_{\text{up}}^{(L_{up})} + \mathbf{I}_{\text{res}}^{(3)}) \tag{7}$$

In the discriminator, the input image $\mathbf{I}$ and the generated image $\mathbf{I}_{\text{out}}$ are first concatenated along the channel dimension, forming the feature map $\mathbf{I}_{\text{FAKE}} \in \mathbb{R}^{H\times W\times 2C}$. Similarly, $\mathbf{I}$ and the ground-truth image $\mathbf{I}_{\text{true}}$ are concatenated to form $\mathbf{I}_{\text{TRUE}} \in \mathbb{R}^{H\times W\times 2C}$. Then, both $\mathbf{I}_{\text{FAKE}}$ and $\mathbf{I}_{\text{TRUE}}$ are applied to four downsampling blocks. In each downsampling block, the spatial dimensions of feature map are reduced by a factor of 1/2 in height and width. Given that $\mathbf{I}_{\text{down}}^{(0)} = \mathbf{I}_{\text{FAKE}}$ or $\mathbf{I}_{\text{down}}^{(0)} = \mathbf{I}_{\text{TRUE}}$, the processing in each block is defined by Eq. (8).

$$\begin{aligned} \mathbf{I}_{\text{down}}^{(l''')} &= \text{Downsample}(\mathbf{I}_{\text{down}}^{(l'''-1)}) \\ l''' &= 1, ..., 4 \end{aligned} \tag{8}$$

$\mathbf{I}_{\text{down}}^{(l''')}$ denotes the output of the l'''th downsampling block. Finally, a convolutional layer is applied, and a score map $\mathbf{M}_{score} \in \mathbb{R}^{H/16\times W/16\times 1}$ is output to

determine whether the input pair is a pair of generated images or a pair of target images. The processing in the last layer is defined by Eq. (9).

$$\mathbf{M}_{score} = \mathrm{Conv}(\mathbf{I}^{(4)}_{down}) \tag{9}$$

2.3 Loss function

pix2pix introduces an adversarial loss L_{GAN} with the discriminator D to learn output images by the generator G to be similar to real data. The adversarial loss is defined by Eq. (10).

$$\begin{aligned} L_{\mathrm{GAN}}(G, D) = \mathbf{E}_{(x,y)}[\log D(x, y)] \\ + \mathbf{E}_{(x,G(x))}[\log(1 - D(x, G(x)))] \end{aligned} \tag{10}$$

where x is the input image, $G(x)$ is the generated image, and y is the target image. $D(x, y)$ denotes the probability of identifying a pair of input and target image as a pair of target image, and $1 - D(x, G(x))$ denotes the probability of identifying a pair of input and generated image as a pair of generated images.

Instead of a L1 loss between pixels used in pix2pix, we use a perceptual loss [10] based on features extracted by VGG19 [16], the network used for object recognition. The perceptual loss L_{VGG} is the sum of the absolute differences in feature values of the generated and target images input to the intermediate layer of VGG19(relu1_2, relu2_2, relu3_4, relu4_4, relu5_4). Perceptual loss is defined by Eq. (11).

$$L_{\mathrm{VGG}} = \sum_j \|\phi_j(G(x)) - \phi_j(y)\| \tag{11}$$

where $\phi_j(\cdot)$ denotes the jth intermediate layer.

An overall network loss is defined by Eq. (12).

$$L = \min_G \max_D L_{\mathrm{GAN}}(G, D) + \lambda L_{\mathrm{VGG}} \tag{12}$$

where λ denotes the weight of L_{VGG}.

3 Experiment

To verify the effectiveness of the proposed method for generating the appearance of metallic materials based on the wide range of lighting conditions in the input image, we compared the proposed method with a baseline method using CNN-based image-to-image translation network. The CNN-based image-to-image translation network was based on pix2pix, where the generator was a CNN-based U-Net [14] architecture and the discriminator was the same as that used in the proposed method. The loss function of the network was the same as that used in the proposed method.

3.1 Dataset

By creating an environment map based on an omnidirectional image and rendering a scene with 3D objects placed in it, an input image containing a virtual object with the appearance of metallic materials was created. We used Blender, a 3DCG production software, to render the virtual scene, and created 50,000 pairs for training and 10,000 pairs for testing. Fig. 3 shows an overview of the procedure for creating pairs of input images and target images.

We acquired the omnidirectional images from Poly Haven and used 30 images from the Outdoor, Natural Light, High Contrast category that included the high-intensity sun and 30 images from the Outdoor, Natural Light, Low Contrast category that included cloudy days and shaded environments. We selected 25 random images from each category for the training data, for a total of 50 images, and used the remaining 5 images each, for a total of 10 images, as the test data. We used Bunny from The Stanford 3D Scanning Repository [13] as the virtual object.

We set the metallic parameter to 1.0, the roughness parameter to 0.2, and the specular parameter to 1.0 for the material of the 3D object. The camera was randomly placed in the range of radius $r \in [0.25, 0.50]$, elevation angle $\theta \in [0, \frac{3}{4}\pi]$, and azimuth angle $\phi \in [0, 2\pi]$ based on a spherical coordinate system, and set the orientation to face the origin. The camera pose was randomly changed within the specified range for each environment map, and 1,000 images were rendered.

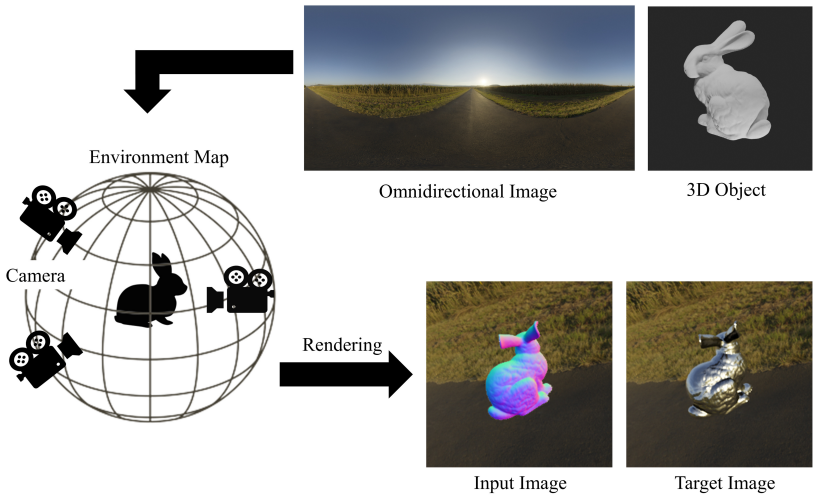

Fig. 3: How to create an input image and a target image

3.2 Training setup

In this experiment, the network was trained with a batch size of 16 at epoch 100. The weight λ in the network loss function Eq. (12) was set to 2.0. Adam was used as the optimization algorithm by setting the learning rate of the generator $\alpha_g = 2 \times 10^{-4}$, the learning rate of the discriminator $\alpha_d = 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. In the generator, the input image was divided into patches of 8×8 pixels, and each patch was projected into the 64 dimensional patch embeddings. In the discriminator, the patch size was set to 16×16. We used Intel(R) Core(TM) i7-13700 CPU and NVIDIA GeForce RTX 4070 GPU as the hardware used for training.

3.3 Experimental results

Using the test data as input, we performed a quantitative evaluation of the generated images by the proposed method with ViT and the baseline method with CNN. The quantitative evaluation values using the three evaluation metrics PSNR, SSIM, and LPIPS [21], are shown in Tab. 1.

Table 1: Quantitative evaluation of generated images in three metrics

Network	PSNR	SSIM	LPIPS
Baseline method	21.28	0.8499	0.1140
Proposed method	**21.67**	**0.8570**	**0.1077**

LPIPS is a metric that evaluates visual similarity by calculating the distance between the feature vectors in the intermediate layers of the generated and reference images using a trained CNN. This metric gives results closer to those obtained by human vision. In this experiment, we used the trained model of AlexNet [12] for the calculation of LPIPS. The results shown in Tab. 1 demonstrate that the proposed method performed better in all metrics.

To confirm whether the proposed method can generate the appearance of metallic materialsbased on the long-range dependencies of light in the input image, we compared the generated images by the proposed method using ViT and the baseline method using CNN.

Fig. 4 shows the generated images using the proposed method and the baseline method with the test data. The images in the first row were captured in shaded areas, while those in the second row were captured in locations exposed to direct sunlight. The third and fourth rows show generated images using a high-contrast image as the input for the environment map, in which the sun was located behind the virtual object. Although the generated images differed from the ground truth, both methods successfully reproduced natural reflections and highlights in the first and second rows. In the third and fourth rows, the baseline method generated unnatural highlights on the top and the bright surface

(a) Input | (b) Baseline method (CNN) | (c) Proposed method (ViT) | (d) Ground Truth

Fig. 4: Example results using the baseline method (CNN) and the proposed method (ViT)

of the virtual object. This is considered to be caused by the strong influence of high-luminance regions and excessive brightness enhancement without adequately reflecting the extensive light of the background in the output image. On the other hand, the proposed method generated images in which the surface of the virtual object was dimly lit, as in the ground truth, successfully reproducing backlighting. This suggests that the proposed method learned the long-range dependencies in the entire input image and more accurately reflected the lighting information in the output image. As a result, the proposed method reproduced the visually natural appearance of the metallic material, even though there were

slight differences from the ground truth in the background reflected on the surface.

4 Conclusion

In this study, we proposed a method for generating images that reproduce the appearance of metallic materials by using an image-to-image translation network with ViT. The introduction of ViT, which effectively captures long-range dependencies in the entire input image, enhanced the network's ability to more accurately reflect the estimated lighting information in the output image. As a result, the network achieved robust performance under lighting conditions where high-intensity light sources were present in the background, and successfully reproduced the appearance of metallic materials consistent with the surrounding environment of the virtual object.

In future work, since we trained the proposed method on a dataset with images of outdoor environments as the background, it is not clear whether the proposed method can adapt to input images where the background is an indoor environment. A remaining challenge is to develop a general-purpose image-to-image translation network capable of reproducing the appearance of metallic materials in both indoor and outdoor environments by extracting relevant features from the input images to generate optimal reflections tailored to each environment.

References

1. Chen, H., Wang, Y., Guo, T., Xu, C., Deng, Y., Liu, Z., Ma, S., Xu, C., Xu, C., Gao, W.: Pre-trained image processing transformer. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 12299–12310 (2021)
2. Debevec, P.: Rendering synthetic objects into real scenes: bridging traditional and image-based graphics with global illumination and high dynamic range photography. In: Proceedings of the Annual Conference on Computer Graphics and Interactive Techniques. pp. 189–198 (1998)
3. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. In: Proceedings of the International Conference on Learning Representations (2021)
4. Frahm, J.M., Koeser, K., Grest, D., Koch, R.: Markerless augmented reality with light source estimation for direct illumination. In: Proceedings of the Conference on Visual Media Production. pp. 211–220 (2005)
5. Gardner, M.A., Sunkavalli, K., Yumer, E., Shen, X., Gambaretto, E., Gagné, C., Lalonde, J.F.: Learning to predict indoor illumination from a single image. arXiv preprint arXiv:1704.00090 (2017)
6. Gündüç, Y.: Vit-gan: Image-to-image translation with vision transformes and conditional gans. arXiv preprint arXiv:2110.09305 (2021)
7. Hold-Geoffroy, Y., Sunkavalli, K., Hadap, S., Gambaretto, E., Lalonde, J.F.: Deep outdoor illumination estimation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 7312–7321 (2017)

8. Isola, P., Zhu, J.Y., Zhou, T., Efros, A.A.: Image-to-image translation with conditional adversarial networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1125–1134 (2017)
9. Jiang, Y., Chang, S., Wang, Z.: Transgan: Two transformers can make one strong gan. arXiv preprint arXiv:2102.07074 (2021)
10. Johnson, J., Alahi, A., Fei-Fei, L.: Perceptual losses for real-time style transfer and super-resolution. In: Proceedings of the Computer Vision – ECCV 2016. pp. 694–711. Springer (2016)
11. Kanbara, M., Yokoya, N.: Geometric and photometric registration for real-time augmented reality. In: Proceedings of the IEEE International Symposium on Mixed and Augmented Reality. pp. 279–280 (2002)
12. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. In: Proceedings of the Advances in Neural Information Processing Systems. pp. 1097–1105 (2012)
13. Levoy, M., Gerth, J., Curless, B., Pulli, K.: The stanford 3d scanning repository. https://graphics.stanford.edu/data/3Dscanrep/ (2005)
14. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. arXiv preprint arXiv:1505.04597 (2015)
15. Sato, I., Sato, Y., Ikeuchi, K.: Acquiring a radiance distribution to superimpose virtual objects onto a real scene. IEEE Transactions on Visualization and Computer Graphics **5**(1), 1–12 (1999)
16. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556 (2014)
17. Thomas, M.M., Forbes, A.G.: Deep illumination: Approximating dynamic global illumination with generative adversarial network. arXiv preprint arXiv:1710.09834 (2017)
18. Unger, J., Gustavson, S., Ynnerman, A.: Densely sampled light probe sequences for spatially variant image based lighting. In: Proceedings of the 4th International Conference on Computer Graphics and Interactive Techniques in Australasia and Southeast Asia. pp. 341–347 (2006)
19. Wang, X., Wang, K., Lian, S.: Deep consistent illumination in augmented reality. In: Proceedings of the IEEE International Symposium on Mixed and Augmented Reality Adjunct. pp. 189–194 (2019)
20. Zhan, F., Zhang, C., Yu, Y., Chang, Y., Lu, S., Ma, F., Xie, X.: Emlight: Lighting estimation via spherical distribution approximation. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 35, pp. 3287–3295 (2021)
21. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The unreasonable effectiveness of deep features as a perceptual metric. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 586–595 (2018)